2A Outline: Utility and Choice

I. Definitions and Assumptions

A. Definitions

1. economics - the study of "the ordinary business of life"; the study of the allocation of scarce resources among alternative uses; **"limited resources, unlimited wants"** (memorize this)

2. Utility – a measure of how satisfied a person is

   a. Cardinal Utility – the ability to attach a specific number of utils to a consumption bundle of goods

   b. Ordinal Utility – the ranking of utility received from consuming a bundle of goods

   c. Total Utility – the aggregate amount of satisfaction received from consuming a good

   d. Marginal Utility – MU- the additional satisfaction derived from consuming an additional amount of a good. MU = ΔTU/ΔQ

   e. Law of Diminishing MU – the principle that each additional unit of a good eventually provides less and less extra utility

   Example:

Movies per month (Q)	Total Utility	Marginal Utility (MU = ΔTU/ΔQ)
0	0	
1	50	
2	90	
3	120	
4	140	

   f. Utility function – U = f(X,Y) – a mathematical function that describes the relationship between what consumers actually consume and their level of satisfaction

B. Assumptions about preferences (we will relax these assumptions later)

1. Preferences are complete (and rankable)

2. Preferences are transitive

3. More of a good is preferred to less

4. Preferences do not depend on income

5. Preferences do not depend on prices

Be sure you know this

II. Indifference Curves – a graphical way of representing preferences (*"**unlimited wants**"*)

A. The basic model: Example: Assume John has 3 sodas and 4 burgers

HINT: Focus on the process of creating the indifference curve

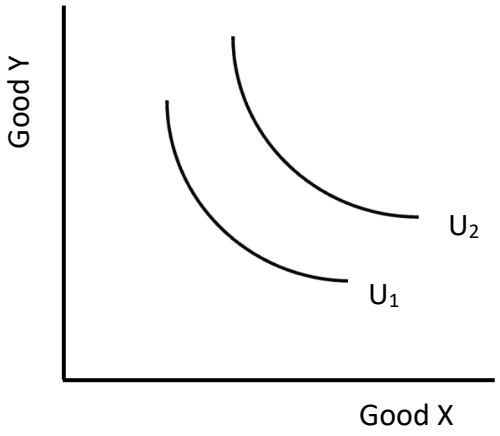
1. The typical shape



Note: We put "things that bring you utility/satisfaction" on the axes

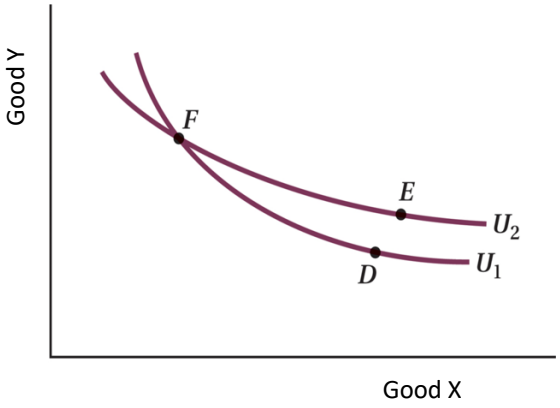
Note: Points along an indifference curve provide the same utility to the consumer, they are all equally preferred

2. Indifference map



Note: All points on U_2 provide more utility than all points on U_1, so $U_2 > U_1$

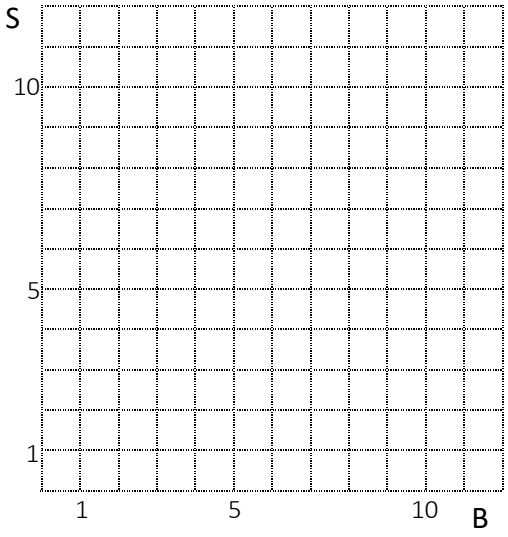
3. Indifference curves cannot cross



B. Marginal Rate of Substitution – the rate at which an individual is willing to reduce consumption of one good when he or she gets one more unit of another good (and still maintain the same level of utility). It is the negative of the slope of an indifference curve. **$MRS_{x,y} = -\Delta Y/\Delta X$**

1. Example. Suppose John receives utility from consuming soda and burgers and that his utility function is given by: **U = SB.**

a. Derive the indifference curve for U = 12. Hint: the combination 1 burger and 12 sodas is one point on his indifference curve.



Written in (X,Y) format

b. Calculate the $MRS_{x,y}$ between the points (2,6) and (3,4) on the indifference curve.

c. Calculate the $MRS_{x,y}$ at the point (2,6) on the indifference curve by drawing a tangent line.

(We will complete parts d and e after discussing material on the next page)

d. Now use $MRS_{x,y} = MUx/MUy$ to calculate the $MRS_{x,y}$ at the point (2,6).

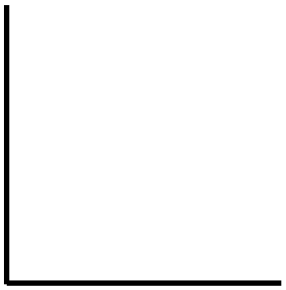
i. Calculate MU_x

ii. Calculate MU_y

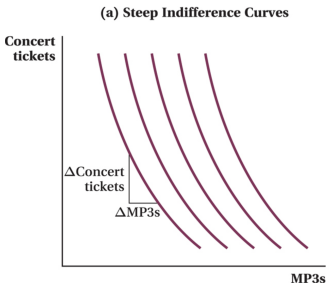
iii. Calculate $MRS_{x,y}$ at (2,6)

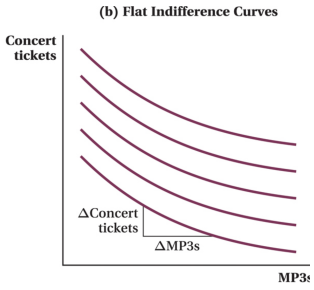
e. Suppose instead that total utility is given by: $U = X^2Y$. Calculate the $MRS_{x,y}$ (an equation)

2. Diminishing marginal rate of substitution and *balance in consumption*



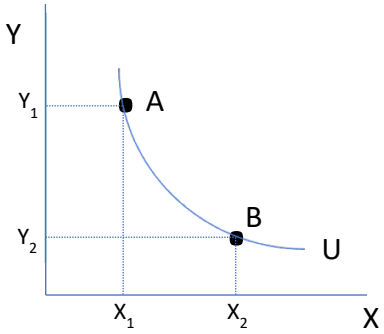
3. The meaning behind the steepness of indifference curves





4. Derivation: $MRS_{x,y} = MUx/MUy$

a. **The Intuition** (p. 116 – 117 of the textbook)



Recall: $MU_X = \Delta TU/\Delta X$

$MRS_{x,y} = -\Delta Y/\Delta X$

If we move from point A to point B:

$\Delta TU = 0$	(Total Utility does not change along an indifference curve)
$MU_X * \Delta X$	Represents the utility gained by consuming more X
$MU_Y * \Delta Y$	Represents the utility lost by consuming less Y

Together, $\Delta TU = MU_X * \Delta X + MU_Y * \Delta Y = 0$

Rearranging, $MU_x/MU_Y = -\Delta Y/\Delta X$

Substituting, $MRS_{x,y} = MU_x/MU_Y$

Note: Now complete the rest of the example on the previous page.

b. **The Calculus** (p. 148 of the textbook) You will not be responsible for this derivation.

$$U(X,Y) = \overline{U}$$

We are interested in how the utility from each good changes as we change the quantities of X and Y. So, we will totally differentiate the utility function, setting the total change in utility, dU, equal to zero because we are holding the level of utility constant:

$$dU = \frac{\partial U(X,Y)}{\partial X}\,dX + \frac{\partial U(X,Y)}{\partial Y}\,dY = 0$$

$$\frac{\partial U(X,Y)}{\partial X}\,dX = -\frac{\partial U(X,Y)}{\partial Y}\,dY$$

$$MU_X dX = -MU_Y dY$$

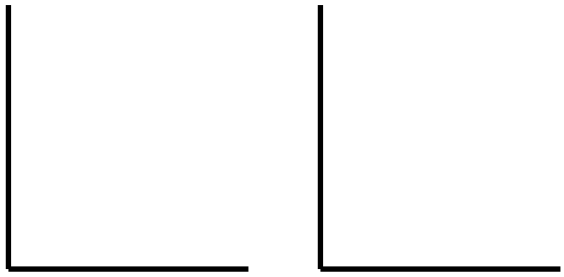
$$\frac{MU_X}{MU_Y} = -\frac{dY}{dX}$$

The right-hand side of this equation, $-dY/dX$, is the negative of the slope of the indifference curve or the marginal rate of substitution. Therefore,

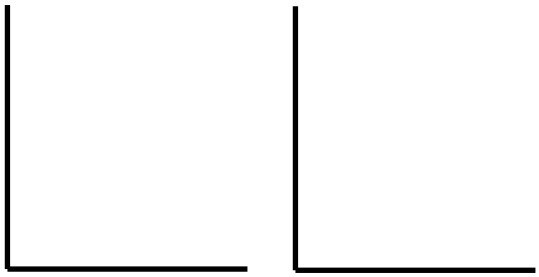
$$MRS_{XY} = \frac{MU_X}{MU_Y}$$

C. **<u>Special Cases</u>** – draw an indifference map for each of the following. Draw and label at least two indifference curves to reflect the fact $U_2>U_1$. Follow the same process we used on page 2 of the handout.

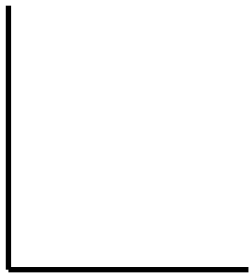
1. A useless good – a widget



2. An economic bad – ants at a picnic



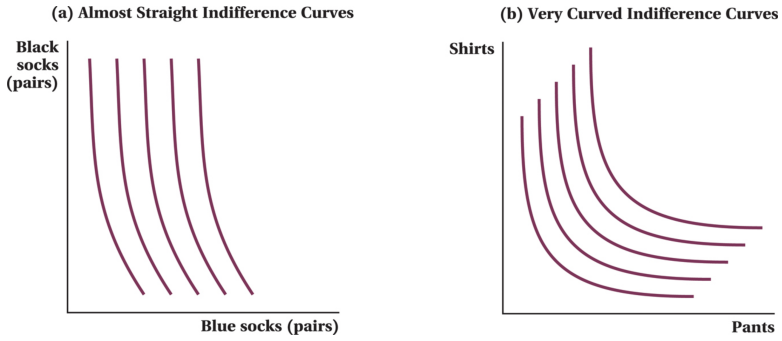
3. Perfect substitutes – brands of gasoline



4. Perfect complements – left and right shoes



D. The curvature of indifference curves



Memorize:

$MU_X = \Delta TU/\Delta X = \left(\frac{\partial TU}{\partial x}\right)$

$MU_Y = \Delta TU/\Delta Y = \left(\frac{\partial TU}{\partial y}\right)$

$MRS_{x,y} = -\Delta Y/\Delta X$

$MRS_{x,y} = MU_x/MU_Y$